

1. Administrative & Study Identification

Full Study Title: AN INTERVENTIONAL EFFICACY AND SAFETY, PHASE 3, RANDOMIZED, DOUBLE-BLIND, 3-ARM STUDY TO INVESTIGATE IBUZATRELVIR IN ADULTS WITH SYMPTOMATIC COVID-19 WHO ARE SEVERELY IMMUNOCOMPROMISED **Short Title/Acronym:** A Study to Learn About the Study Medicine Ibuzatrelvir in Adults With COVID-19 Who Are Severely Immunocompromised **Protocol Number:** C5091018 **NCT Number:** NCT07013474 **Posting ID:** 309128768336694799 **Sponsor/Company Name:** Pfizer **Investigational Product:** Ibuzatrelvir (oral) and Remdesivir (intravenous) **Phase of Development:** Phase 3 **Trial Status:** RECRUITING **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of ibuzatrelvir alone and in combination with remdesivir IV compared to remdesivir IV alone for the treatment of symptomatic COVID-19 in severely immunocompromised adult participants who are non-hospitalized or are hospitalized for observation but do not require supplemental oxygen for COVID-19. **Secondary Objectives:** To evaluate the time to sustained alleviation of all targeted COVID-19 symptoms, to monitor the evidence of recurrence or persistence of SARS-CoV-2 infection, and to assess COVID-19-related ED visits, hospitalizations, or all-cause mortality.

2.2 Background & Rationale Immunocompromised patients with COVID-19 have more difficulty fighting infections and are at high risk for persistent infections and progression to severe illness. This patient population may uniquely benefit from extended antiviral treatment durations or a combination of antiviral therapies (such as a protease inhibitor like ibuzatrelvir combined with a nucleotide analog like remdesivir) to promote sustained viral clearance and lower the risk of viral rebound.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active-controlled, Placebo-controlled **Blinding:** Double-blinded, double-dummy **Randomization:** Randomized 1:1:1 into three arms (Ibuzatrelvir + Remdesivir placebo; Remdesivir IV + Ibuzatrelvir placebo; or Ibuzatrelvir + Remdesivir IV).

3.2 Planned Enrollment & Duration Target \$\$\$: 300 participants
Number of Sites: Approximately 117 locations globally **Estimated Study Start:** 14-Jul-2025 **Estimated Primary Completion:** 06-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. 18 years of age or older at screening. 2. Confirmed SARS-CoV-2 infection as determined by RAT (or other locally approved test) collected within 2 days prior to randomization. Initial onset of symptoms attributable to COVID-19 within 5 days prior to the day of randomization and at least 1 of the specified symptoms attributable to COVID-19 present on the day of randomization. 3. Severely immunocompromised due to: - Solid organ or islet cell transplant recipient who is receiving immunosuppressive therapy; - Active hematologic malignancy (eg, chronic lymphocytic lymphoma, non-Hodgkin lymphoma, multiple myeloma, acute leukemia); - Receipt of CAR-T-cell therapy or HCT either within 2 years of transplantation or who are receiving immunosuppressive therapy; - Currently receiving B-cell depleting therapies (eg, rituximab).

4.2 Exclusion Criteria 1. Current need for supplemental oxygen for treatment of COVID-19. 2. Receiving dialysis or have known severe renal impairment within 6 months of the screening visit. 3. Active liver disease. 4. History of hypersensitivity or other contraindication to any of the components of the study interventions, as determined by the investigator. 5. Suspected or confirmed concurrent active systemic infection other than COVID-19 that may interfere with the evaluation of response to the study intervention. 6. Life expectancy less than 30 days at study entry due to an underlying condition, in the judgement of the investigator. 7. Any medical or psychiatric condition including recent (within the past year) or active suicidal ideation/behavior or laboratory abnormality that may increase the risk of study participation or, in the investigator's judgment, make the participant inappropriate for the study. 8. Has received any other antiviral for the treatment of the current COVID-19 infection. 9. Current use of any prohibited concomitant medication(s) or unwillingness or inability to use a required concomitant medication(s). 10. Current or previous administration of an investigational product (drug or vaccine) within 30 days (or as determined by local requirement) or 5 half lives preceding the first dose of study intervention used in this study (whichever is longer). Authorized or products with conditional approval are not considered investigational. 11. Prior participation in this trial or any clinical trial of ibuzatrelvir. 12. Females who are pregnant, breastfeeding, or who are planning to become pregnant within the timeframe of the study. 13. Investigator site staff directly involved in the conduct of the study and their family members, site staff otherwise supervised by the investigator, and sponsor and sponsor

delegate employees directly involved in the conduct of the study and their family members.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ibuzatrelvir taken by mouth twice daily; Remdesivir (Trade Name: Veklury, ATC Code: J05AB16) administered as an IV infusion daily. Placebos matching both study medicines are provided to maintain the double-dummy blinding across all three treatment arms. **Route of Administration:** Oral (Ibuzatrelvir/Placebo) and Intravenous (Remdesivir/Placebo). **Duration of Treatment:** Specified extended duration for severely immunocompromised individuals per randomization arm protocol.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for underlying severe immunocompromising conditions not known to dangerously interact with investigational products. **Prohibited:** Receipt of any other antiviral for the treatment of the current COVID-19 infection or use of strongly contraindicated concomitant medications.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -2 to 0 | Informed Consent, eligibility check, COVID-19 RAT confirmation. | | **Baseline** | Day 0 | Randomization, First Dose (IV and Oral), symptom questionnaire baseline. | | **Interim** | Over 24 Weeks | Approximately 10 clinic visits: Blood work, clinic nasal swabs, dispensing of at-home nasal swab kits, and symptom questionnaires. | | **End of Study** | Week 24 | Final safety blood work, adverse event tracking, overall outcome assessment, and exit interview. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The difference in the proportion of participants with the composite endpoint of a) COVID-19-related ED visits with treatment, COVID-19-related hospitalization, or all-cause mortality, and b) evidence of recurrent or persistent SARS-CoV-2 infection: The difference in the proportion of patients meeting the primary composite endpoint, between ibuzatrelvir and remdesivir vs remdesivir groups in symptomatic adult participants with COVID-19 who are severely immunocompromised.

7.2 Secondary & Exploratory Endpoints 1. Time to sustained alleviation of all targeted COVID-19 symptoms. 2. Proportion of participants with evidence of recurrent or persistent SARS-CoV-2

infection. 3. Proportion of participants with COVID-19-related ED visits with administration of supplemental oxygen, COVID-19 antiviral, or IV treatment, COVID-19-related hospitalization, or all-cause mortality.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via patient-reported questionnaires, at-home symptom tracking, and lab evaluations during the ~10 clinic visits over 24 weeks. **Serious Adverse Events (SAEs):** Standard reporting timelines strictly enforced (typically 24 hours). **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (DMC) evaluates safety and scientific integrity throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary superiority outcomes. Safety Set for evaluating AE/SAEs across the three arms. **Sample Size Justification:** Target enrollment of $N=300$ participants properly powers the detection of superiority between the active combination arm against the active monotherapy arm. **Handling of Missing Data:** Standardized imputation for missing participant-level data (IPD) regarding daily symptom relief and viral load quantifications.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Monitoring across 117 global locations to ensure proper blinding and double-dummy administration compliance. **Audit Trail:** Robust electronic diary (e-diary) tracking for continuous daily symptom reporting and proper eCRF lock procedures.

11. Study Identifiers & Governance

Primary ID: C5091018 **Registry Number:** NCT07013474 **Posting ID:** 309128768336694799 **Sponsor/Lead Organization:** Pfizer **Study Title:** A Study to Learn About the Study Medicine Ibuzatrelvir in Adults With COVID-19 Who Are Severely Immunocompromised **Official Regulatory Title:** AN INTERVENTIONAL EFFICACY AND SAFETY, PHASE 3, RANDOMIZED, DOUBLE-BLIND, 3-ARM STUDY TO INVESTIGATE IBUZATRELVIR IN ADULTS WITH SYMPTOMATIC COVID-19 WHO ARE SEVERELY IMMUNOCOMPROMISED

12. Clinical Framework (COVID-19 Specific)

Disease Area: COVID-19 Infection **Condition Name:** COVID-19
(Preferred Name: COVID-19 Virus Disease / Preferred UMLS Name: COVID-19 in immunocompromised host) **Semantic Type:** Disease or Syndrome **Definition:** An acute infection of the respiratory tract that is caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). **MeSH Code:** D000086382 **SNOMED ID:** 840539006 **Diagnosis Threshold:** Confirmed infection via RAT within 2 days with ≤ 5 days of symptom onset. **Medical Methodology:** Extended combination antiviral therapy **Optimization Target:** Prevention of persistent infection, reduction of viral rebound, and sustained symptom resolution

13. Study Procedures & Methodology

Management Pathway: Outpatient monitoring or hospitalization purely for observation/intervention (no supplemental oxygen requirement). **Education Protocol (HCP):** Site instruction for maintaining double-dummy blinding matching oral and IV placebos. **Education Protocol (Patient):** Education on capturing swabs at home, medication adherence, and electronic questionnaire usage. **Quality Audit Mechanism:** Real-time data surveillance of at-home nasal swab logs and symptom tracking inputs.

14. Observation & Follow-Up Schedule

Total Duration: 24 weeks **Onsite Visit Cadence:** ~10 clinic visits distributed across the 24-week period **Remote Monitoring Frequency:** Continuous patient-reported outcome tracking (questionnaires) **Checklist Review Interval:** Routine follow-up per clinic visit schedule.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				